Antacids

Medical uses (indications)

* Heartburn (Gastro-oesophageal/ Gastroesophageal reflux)

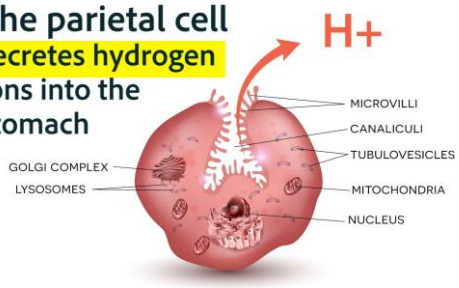
Contraindications

* GORD/GERD (The chronic disorder)



1

The parietal cell secretes hydrogen ions into the stomach



2

Antacids

H+ and Cl-

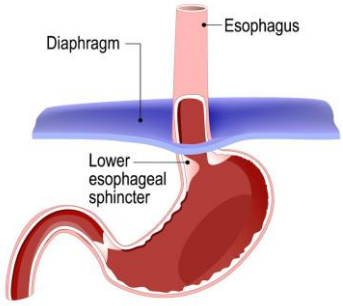
= HCl (hydrochloric acid)

H_2O (Water)

3

The normal stomach

Note the location of the sphincter



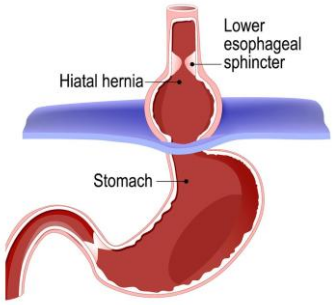
4

Sphincter

Has some 'tone'

Affected by:

* Drugs
* 'Full' stomach
* Hiatal hernia



5

Safety issues associated with antacids

Chronic insults to tissues must be assessed for further damage

The correct medication must be prescribed

6

The 'rebound' effect of many medications



Rebound hyperacidity

7

The 'rebound' effect of many medications

The body attempts to establish homeostasis (equilibrium)



8

The acid in the stomach

* Helps absorption of some nutrients
* Helps absorption of many drugs

Using that information, predict the best time to take antacids

Answer: One to two hours before or after meals or medication

9

Antacids

Magnesium hydroxide (or citrate)

Aluminium (aluminum) hydroxide (or citrate)

Sodium bicarbonate is absorbed

Calcium carbonate is absorbed

Extreme rebound hyperacidity

10



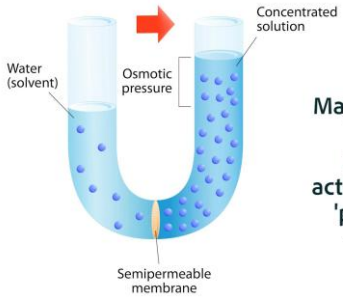
Hydrochloric acid (HCl)
2 (H+ and Cl-)

Magnesium hydroxide
$Mg(OH)_2$

Magnesium chloride
$Mg(Cl)_2$

2 Water
2 (H_20)

11



Magnesium chloride
$Mg(Cl)_2$
is 'osmotically active' (simplistically, 'pulls' water into the intestines)

12

Magnesium antacids often cause diarrhoea



13

Aluminium (Aluminum) Antacids



* Aluminium chloride causes constipation
* Aluminium chloride binds many drugs

Take 2 hours before or after medications

14

Understanding Medications

We are taking our easiest medications first so that we can apply the information to other medications

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Understanding Medications

The laxatives

* Indicated for the treatment of constipation
* Not the 'first line' therapy

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Aims of Chapter 1 (Part B)

* Continue developing the habit of learning in the most productive way
* Understand and be able to describe the safety issues associated with constipation
* Understand and describe the indications, contraindications and side effects of the:
    * Osmotic laxatives
    * Stimulant laxatives
    * Fecal softeners

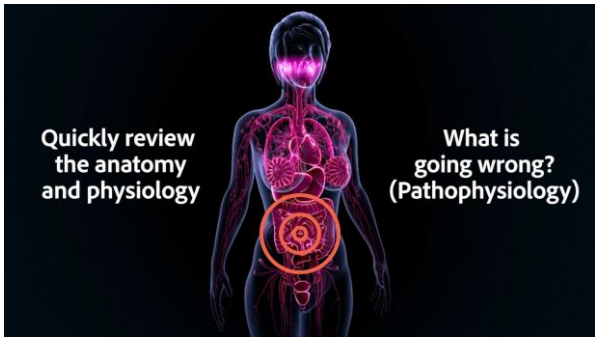
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Definition of constipation

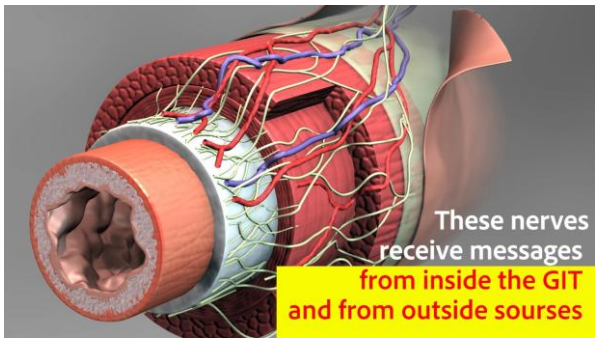
2 or more of the following in 12 weeks

1. Fewer than 3 bowel movements per week
2. Hard stools
3. A sense of incomplete evacuation
4. Excessive straining
5. The need for digital manipulation

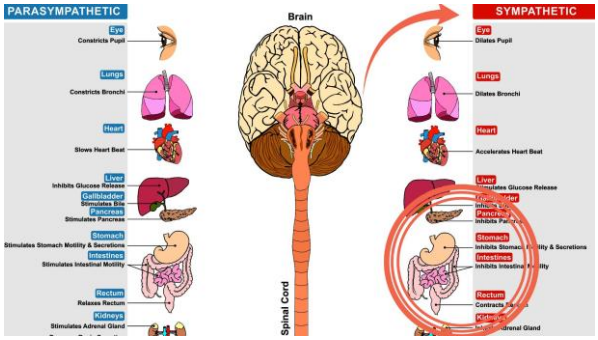
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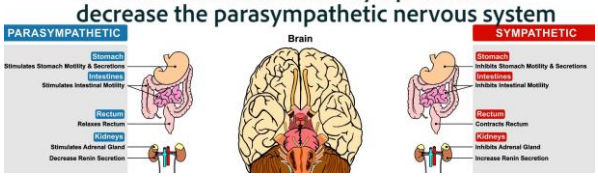
21



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Causes of constipation

* Some are more susceptible to the risk factors
* Medications
  * The ones that increase the sympathetic or decrease the parasympathetic nervous system



23

Causes of constipation

* Some are more susceptible to the risk factors
* Medications
  * The ones that increase the sympathetic or decrease the parasympathetic nervous system
  * The opioids decrease movement
* The contents of the GIT
  * Too little fiber
  * Too little water

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Safety issues concerning constipation

Constipation can lead to:

* Hemorrhoids
* Anal fissures
* Decreased excretion of toxins
* Intestinal blockage

25



26



27

The most important symptoms in intestinal blockage

1. Colicky pain (comes and goes in a wavelike pattern)
2. Small volume mucus-like diarrhea
3. Significant abdominal distension
   (The GIT contents 'backing up')

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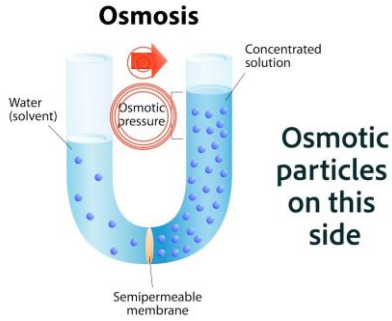
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Osmotic laxatives

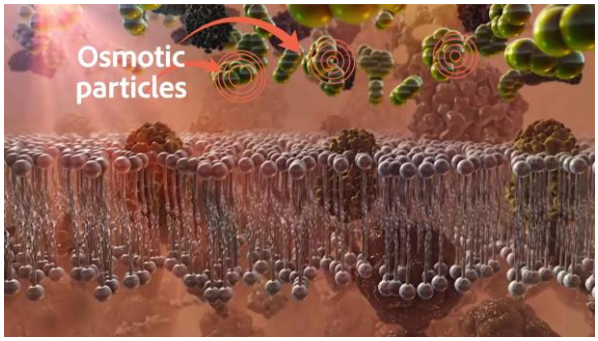
Magnesium citrate (or hydroxide)

Lactulose (a non-absorbable sugar)

30


Osmosis
Concentrated solution
Water (solvent)
Osmotic pressure
Semipermeable membrane
Osmotic particles on this side

31


Osmotic particles

32

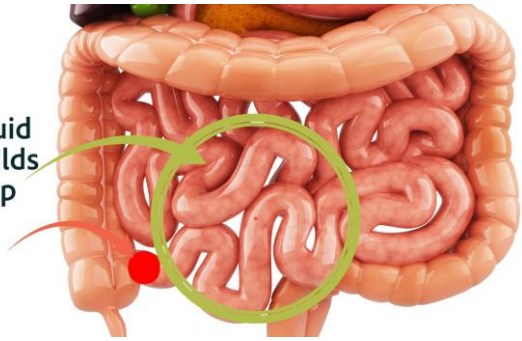

Lumen of the GIT after osmotic laxatives

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Osmotic laxatives are contraindicated in obstruction



34



35

Oral lactulose

A sugar that is not absorbed (it stays in the lumen of the GIT)

Very sweet, so it may need to be diluted

36

Osmotic laxatives



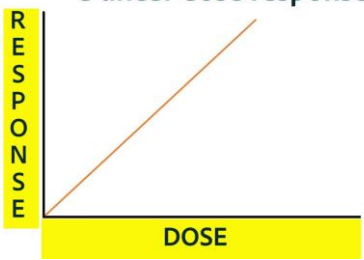
Magnesium citrate (or hydroxide)

(Example: Milk of Magnesia)

Results in magnesium chloride, an osmotically active particle

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Osmotic laxatives demonstrate a linear dose response



38

Many elderly individuals already do not drink adequate water

Disorientation

Dehydration

Dizzy

Falling

39

Water

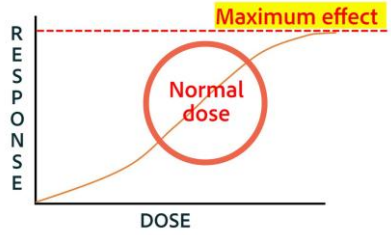
Water

Treatment of constipation

Water

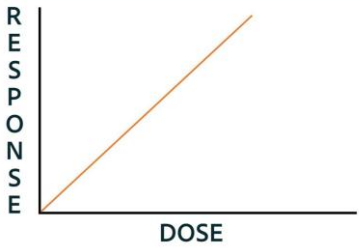
40

Drugs normally demonstrate this type of dose response



41

This type of response is RARE



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Stimulant laxatives

* Bisacodyl
* Sodium picosulfate
* Senna
* Castor oil

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Stimulant laxatives

Very effective / Many side effects

Indicated in OCCASIONAL INFREQUENT instances:

* When nothing else is working
* As a preparation for endoscopy or surgical procedures

44

Stimulant laxatives: Side effects

1. Strong rebound effect (more constipation)

Disrupting the balance often results in this type of effect



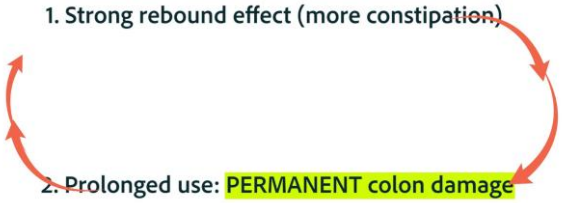
45

Stimulant laxatives: Side effects

1. Strong rebound effect (more constipation)
2. Prolonged use: PERMANENT colon damage
3. Electrolyte abnormalities

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1. Strong rebound effect (more constipation)



2. Prolonged use: PERMANENT colon damage

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Stimulant laxatives should be used infrequently



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Sodium picosulfate

A 'pro-drug' (needs to be metabolized to the active form)

Intestinal bacteria

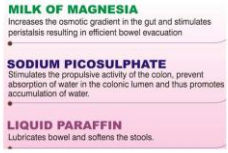


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Sodium picosulfate

A 'pro-drug' (needs to be metabolized to the active form)

Often combined with an osmotic laxative



50

Bisocodyl

* A stimulant laxative
* Not to be used on a regular basis
* May result in proctitis (inflammation of the inner rectum)

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Stimulant laxatives

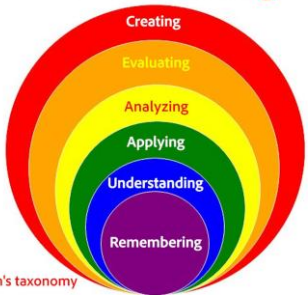
Very effective / Many side effects

Indicated in OCCASIONAL INFREQUENT instances:

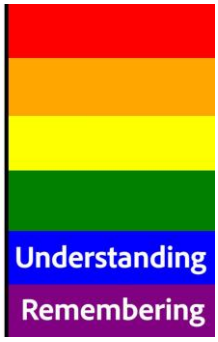
* When nothing else is working
* As a preparation for endoscopy or surgical procedures

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Levels of learning



Bloom's taxonomy



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Which of the following is effective only if taken orally?

* Bisacodyl
* Sodium picosulfate is oral only
* Senna
* Castor oil

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Stimulant laxatives

Bisacodyl, senna and castor oil

Oral: 6-15 hours

Supporitory: 15 minutes to 1 hour

Enema: 5-15 minutes

55

Stimulant laxatives

Bisacodyl, senna and castor oil

Oral: 6-15 hours

Supporitory: 15 minutes to 1 hour

Enema: 5-15 minutes

Sodium picasulfate is only effective if taken orally

56

Enteric coated (delayed release)

This is omeprazole, which is rapidly degraded in stomach acids



* Irritating to stomach
* Rapidly degraded in stomach acids...

57

Mineral oil:

Side effects and safety issues

* Short term, intermittent use only
* Will decrease absorption of fat soluble vitamins
  * Vitamin A
  * Vitamin D
  * Vitamin E
  * Vitamin K

Take on an empty stomach!

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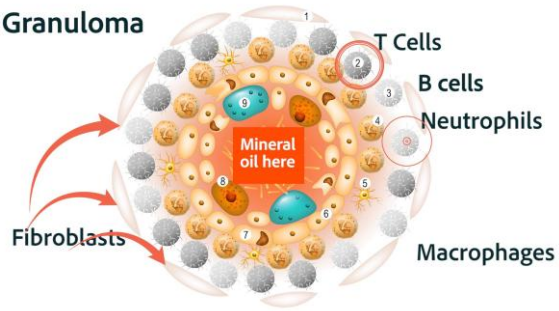
Docusate

(docusate salts; dioctyl sulfosuccinate; fecal softeners; emulsifying laxatives; surfactants)

* Detergent properties soften stools
  * Softens stools but may not relieve constipation
* Absorption of many drugs is affected
  * Toxicity is possible

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Granuloma



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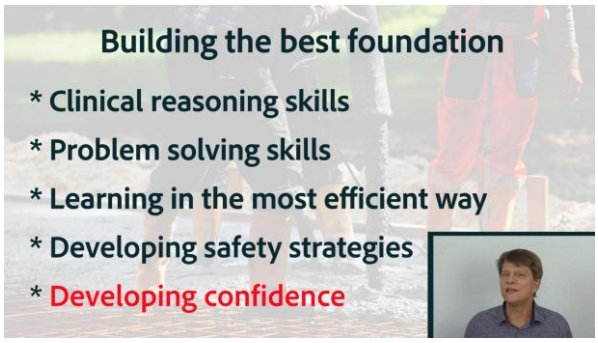
Treatment for constipation

* Mineral oil (normally stays in the intestines)
is strictly contraindicated
in people taking
* Fecal softeners (emulsifies fats)

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Even OTC (over the counter)
medications can be dangerous
if not taken as directed

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References

- Dr Ken Wojcikowski
- PharmacoPhoto
- Royalty free images provided by Shutterstock

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